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INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

Administration Système_ TD N° 6

Préparé par : Brian ANTOINE

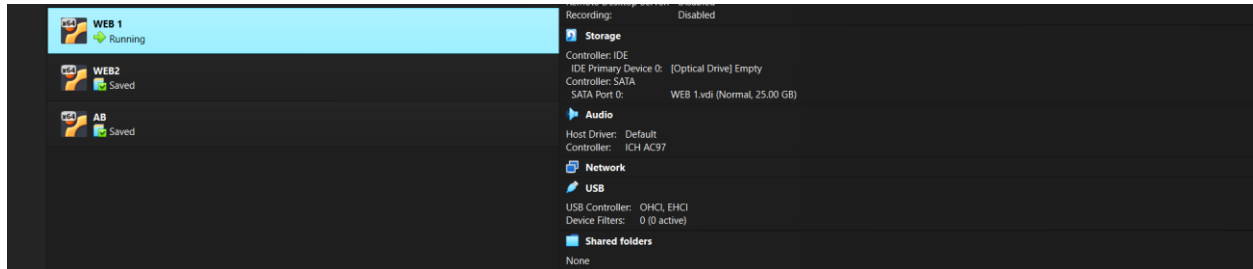
Présenté à: M. Ismaël SAINT AMOUR

Niveau : L3

Date: 12/04/2026

E-commerce HA + Paiement simulé

➤ Machines



1. Installation de Nginx + PHP sur Web1

```
web1@WEB1: ~  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
web1@WEB1:~$ sudo apt update  
[sudo] password for web1:  
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease  
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
30 packages can be upgraded. Run 'apt list --upgradable' to see them.  
web1@WEB1:~$ sudo apt install nginx php-fpm -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  nginx-common php-common php8.3-cli php8.3-common php8.3-fpm php8.3-opcache php8.3-readline  
Suggested packages:  
  fcgiwrap nginx-doc ssl-cert php-pear  
The following NEW packages will be installed:  
  nginx nginx-common php-common php-fpm php8.3-cli php8.3-common php8.3-fpm php8.3-opcache  
  php8.3-readline  
0 upgraded, 9 newly installed, 0 to remove and 30 not upgraded.  
Need to get 5,549 kB of archives.  
After this operation, 24.1 MB of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.
```

```
web1@WEB1:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-13 14:16:20 UTC; 2min 15s ago
     Docs: man:nginx(8)
  Main PID: 6171 (nginx)
    Tasks: 2 (limit: 2263)
   Memory: 1.7M (peak: 3.7M)
      CPU: 66ms
   CGroup: /system.slice/nginx.service
           └─6171 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─6174 "nginx: worker process"

Apr 13 14:16:20 WEB1 systemd[1]: Starting nginx.service - A high performance web server and a re>
Apr 13 14:16:20 WEB1 systemd[1]: Started nginx.service - A high performance web server and a rev>
[lines 1-14/14 (END)]
[1]+  Stopped                  sudo systemctl status nginx
web1@WEB1:~$
```

```
web1@WEB1:~$ systemctl list-unit-files | grep -i php
php8.3-fpm.service          enabled          enabled
phpsessionclean.service    static          -
phpsessionclean.timer      enabled          enabled
web1@WEB1:~$
```

➤ Création du site e-commerce sur Web1

```
web1@WEB1: ~
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
web1@WEB1:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-13 14:16:20 UTC; 2min 15s ago
     Docs: man:nginx(8)
  Main PID: 6171 (nginx)
    Tasks: 2 (limit: 2263)
   Memory: 1.7M (peak: 3.7M)
      CPU: 66ms
   CGroup: /system.slice/nginx.service
           └─6171 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─6174 "nginx: worker process"

Apr 13 14:16:20 WEB1 systemd[1]: Starting nginx.service - A high performance web server and a re>
Apr 13 14:16:20 WEB1 systemd[1]: Started nginx.service - A high performance web server and a rev>
[lines 1-14/14 (END)]
[1]+  Stopped                  sudo systemctl status nginx
web1@WEB1:~$ systemctl list-unit-files | grep -i php
php8.3-fpm.service          enabled          enabled
phpsessionclean.service    static          -
phpsessionclean.timer      enabled          enabled
web1@WEB1:~$ sudo mkdir -p /var/www/shop
web1@WEB1:~$ sudo nano /var/www/shop/index.php
web1@WEB1:~$
```

➤ Permissions

```
web1@WEB1:~$ sudo nano /var/www/shop/index.php
web1@WEB1:~$ sudo chown -R www-data:www-data /var/www/
web1@WEB1:~$ sudo chmod -R 755 /var/www/
web1@WEB1:~$
```

➤ Configuration de Nginx : création du virtual host

```
web1@WEB1: ~  
web1@WEB1:~$ sudo chown -R www-data:www-data /var/www/  
web1@WEB1:~$ sudo chmod -R 755 /var/www/  
web1@WEB1:~$ sudo nano /etc/nginx/sites-available/shop  
web1@WEB1:~$ cat /etc/nginx/sites-available/shop  
cat: /etc/nginx/sites-aivalable/shop: No such file or directory  
web1@WEB1:~$ cat /etc/nginx/sites-available/shop  
server {  
    listen 80;  
    server_name shop.devcloud.local;  
  
    root /var/www/shop;  
    index index.php;  
  
    location / {  
        try_files $uri $uri/ =404;  
    }  
  
    location ~ \.php$ {  
        include snippets/fastcgi-php.conf;  
        fastcgi_pass unix:/run/php/php8.2-fpm.sock;  
    }  
}  
}~server {  
    listen 80;  
    server_name shop.devcloud.local;  
  
    root /var/www/shop;  
    index index.php;  
  
    location / {  
        try_files $uri $uri/ =404;
```

➤ Activer le site

```
web1@WEB1: ~  
    location ~ \.php$ {  
        include snippets/fastcgi-php.conf;  
        fastcgi_pass unix:/run/php/php8.2-fpm.sock;  
    }  
}  
EOF  
web1@WEB1:~$ cat /etc/nginx/sites-available/shop  
server {  
    listen 80;  
    server_name shop.devcloud.local;  
  
    root /var/www/shop;  
    index index.php;  
  
    location / {  
        try_files $uri $uri/ =404;  
    }  
  
    location ~ \.php$ {  
        include snippets/fastcgi-php.conf;  
        fastcgi_pass unix:/run/php/php8.2-fpm.sock;  
    }  
}  
web1@WEB1:~$ sudo ln -s /etc/nginx/sites-available/shop /etc/nginx/sites-enabled/  
web1@WEB1:~$ sudo nginx -t  
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
nginx: configuration file /etc/nginx/nginx.conf test is successful  
web1@WEB1:~$ sudo systemctl restart nginx  
web1@WEB1:~$
```

2. Installation de Nginx + PHP sur Web2

```
web2@WEB2:~$ sudo apt update
[sudo] password for web2:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,892 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,584 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [254 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [344 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,170 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,934 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [681 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,666 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [323 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7,752 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Fetched 11.3 MB in 2min 6s (89.7 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
35 packages can be upgraded. Run 'apt list --upgradable' to see them.
web2@WEB2:~$ sudo apt install nginx php-fpm -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common php-common php8.3-cli php8.3-common php8.3-fpm php8.3-opcache php8.3-readline
Suggested packages:
```

```
web2@WEB2:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-13 15:13:42 UTC; 15min ago
     Docs: man:nginx(8)
    Main PID: 6728 (nginx)
      Tasks: 3 (Limit: 4601)
    Memory: 2.4M (peak: 5.1M)
       CPU: 63ms
    CGroup: /system.slice/nginx.service
            └─6728 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─6731 "nginx: worker process"
                └─6732 "nginx: worker process"

web2@WEB2:~$ systemctl list-unit-files | grep -i php
php8.3-fpm.service           enabled          enabled
phpsessionclean.service     static          -
phpsessionclean.timer       enabled         enabled

web2@WEB2:~$ systemctl status php8.2-fpm
Unit php8.2-fpm.service could not be found.
web2@WEB2:~$ systemctl status php8.3-fpm
● php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php8.3-fpm.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-13 15:14:16 UTC; 17min ago
     Docs: man:php-fpm8.3(8)
  Process: 8299 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/php/8.3/fpm/pool.
 Main PID: 8296 (php-fpm8.3)
    Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0req/sec"
      Tasks: 3 (Limit: 4601)
    Memory: 7.5M (peak: 8.1M)
       CPU: 284ms
```

➤ Création du site e-commerce sur Web2

+

```
web2@WEB2: ~  
php8.3-fpm.service          enabled          enabled  
phpsessionclean.service    static          -  
phpsessionclean.timer      enabled         enabled  
web2@WEB2:~$ systemctl status php8.2-fpm  
Unit php8.2-fpm.service could not be found.  
web2@WEB2:~$ systemctl status php8.3-fpm  
● php8.3-fpm.service - The PHP 8.3 FastCGI Process Manager  
   Loaded: loaded (/usr/lib/systemd/system/php8.3-fpm.service; enabled; preset: enabled)  
   Active: active (running) since Mon 2026-04-13 15:14:16 UTC; 17min ago  
     Docs: man:php-fpm8.3(8)  
  Process: 8299 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/php/8.3/fpm/pool.  
 Main PID: 8296 (php-fpm8.3)  
   Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0req/sec"  
    Tasks: 3 (limit: 4601)  
  Memory: 7.5M (peak: 8.1M)  
     CPU: 284ms  
   CGroup: /system.slice/php8.3-fpm.service  
           └─8296 "php-fpm: master process (/etc/php/8.3/fpm/php-fpm.conf)"  
             └─8297 "php-fpm: pool www"  
               └─8298 "php-fpm: pool www"  
[1]+  Stopped                  systemctl status php8.3-fpm  
web2@WEB2:~$ udo mkdir -p /var/www/shop  
Command 'udo' not found, but can be installed with:  
sudo apt install udo  
web2@WEB2:~$ sudo mkdir -p /var/www/shop  
[sudo] password for web2:  
web2@WEB2:~$ sudo mkdir -p /var/www/shop  
web2@WEB2:~$ sudo nano /var/www/shop/index.php  
web2@WEB2:~$
```

➤ Permissions

```
web2@WEB2:~$ sudo chown -R www-data:www-data /var/www/  
web2@WEB2:~$ sudo chmod -R 755 /var/www/  
web2@WEB2:~$
```

➤ Configuration de Nginx : création du virtual host

```
web2@WEB2:~$ sudo chown -R www-data:www-data /var/www/  
[sudo] password for web2:  
web2@WEB2:~$ sudo chmod -R 755 /var/www/  
web2@WEB2:~$ sudo nano /etc/nginx/sites-available/shop  
web2@WEB2:~$ sudo ln -s /etc/nginx/sites-available/shop /etc/nginx/sites-enabled/  
ln: failed to create symbolic link '/etc/nginx/sites-enabled/shop': File exists  
web2@WEB2:~$ sudo rm -f /etc/nginx/sites-enabled/default  
web2@WEB2:~$ sudo nginx -t  
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
nginx: configuration file /etc/nginx/nginx.conf test is successful  
web2@WEB2:~$ sudo systemctl restart nginx  
web2@WEB2:~$
```

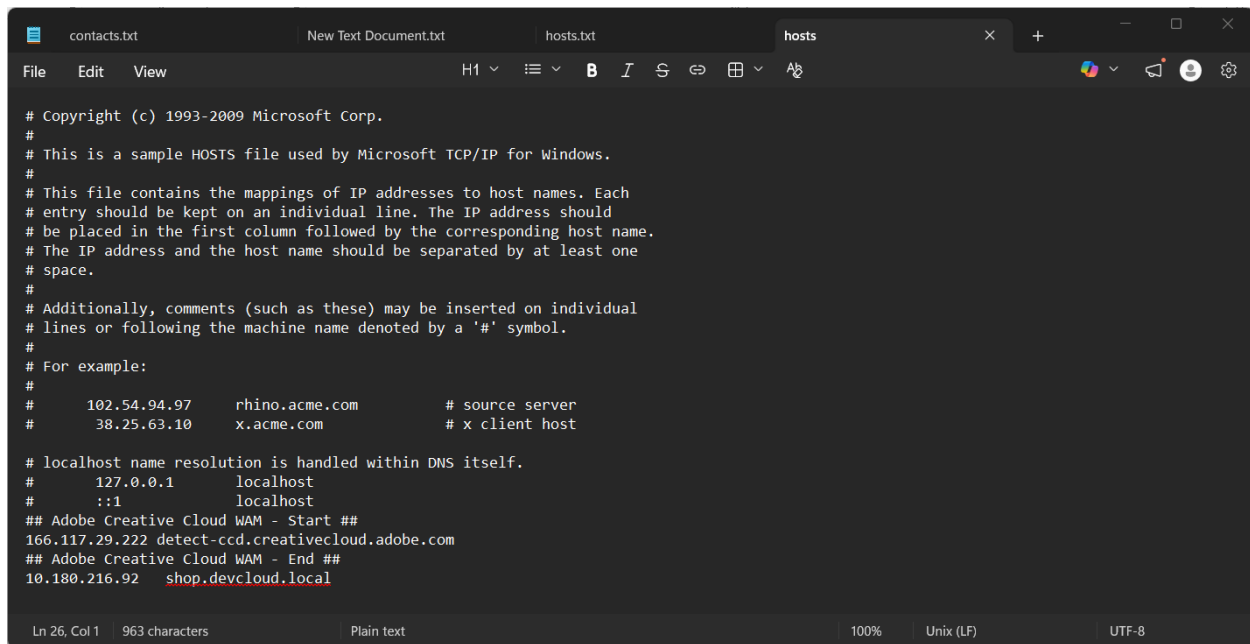
3. Configuration de Load Balancer

➤ J'ai pris mon serveur initial comme load balancer

```
bribri@bribri: ~  
bribri@10.180.216.92's password:  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-107-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Mon Apr 13 10:13:56 PM UTC 2026  
  
System load:  0.03          Processes:            129  
Usage of /:   15.0% of 24.44GB Users logged in:      1  
Memory usage: 7%          IPv4 address for enp0s3: 10.180.216.92  
Swap usage:   0%  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
  just raised the bar for easy, resilient and secure K8s cluster deployment.  
  
https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
Expanded Security Maintenance for Applications is not enabled.  
  
16 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
1 additional security update can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
Last login: Mon Apr 13 17:49:29 2026 from 192.168.2.123  
bribri@bribri:~$
```

```
bribri@bribri: ~  
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
21 packages can be upgraded. Run 'apt list --upgradable' to see them.  
bribri@bribri:~$ sudo apt install nginx -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
nginx is already the newest version (1.24.0-2ubuntu7.6).  
  
bribri@bribri:~$ sudo nginx -t  
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
nginx: configuration file /etc/nginx/nginx.conf test is successful  
bribri@bribri:~$ sudo systemctl restart nginx  
bribri@bribri:~$ sudo systemctl status nginx  
● nginx.service - A high performance web server and a reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)  
   Active: active (running) since Mon 2026-04-13 17:24:01 UTC; 15s ago  
     Docs: man:nginx(8)  
  Process: 1997 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
  Process: 1999 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)  
 Main PID: 2001 (nginx)  
    Tasks: 2 (limit: 4601)  
  Memory: 1.5M (peak: 1.9M)  
     CPU: 35ms  
   CGroup: /system.slice/nginx.service
```

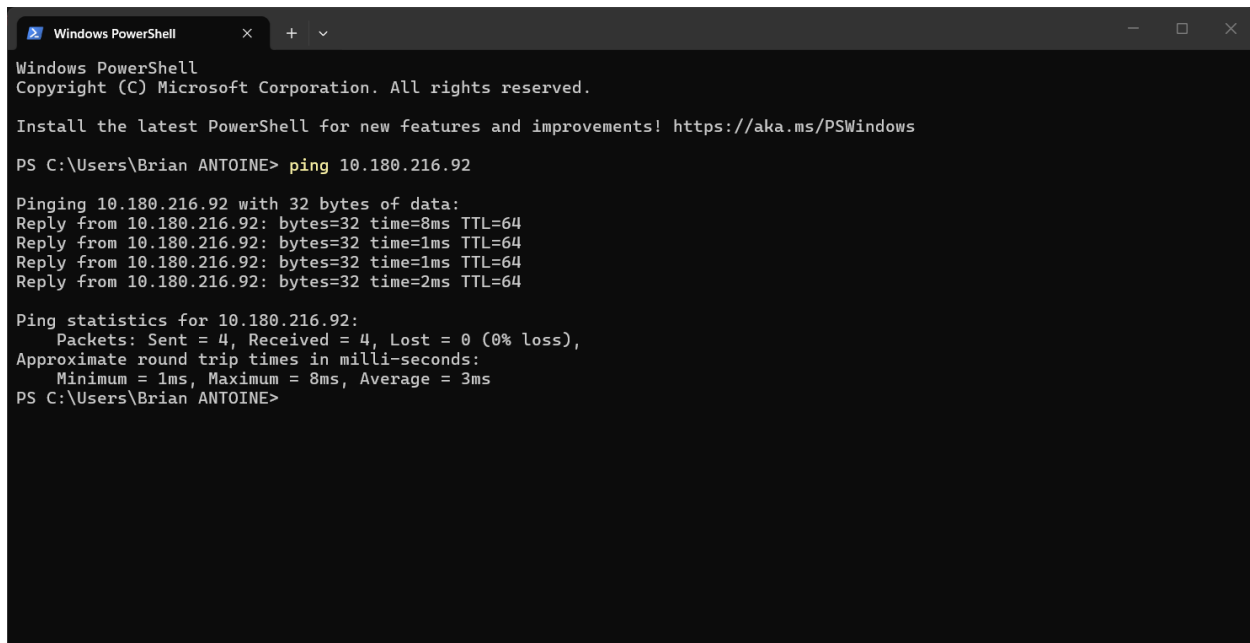
4. Tests depuis Windows



The screenshot shows a text editor window with the file 'hosts.txt' open. The editor has a dark theme and a menu bar with 'File', 'Edit', and 'View'. The text in the file is as follows:

```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com       # source server
#       38.25.63.10       x.acme.com          # x client host
#
# localhost name resolution is handled within DNS itself.
#       127.0.0.1         localhost
#       ::1               localhost
## Adobe Creative Cloud WAM - Start ##
166.117.29.222 detect-ccd.creativecloud.adobe.com
## Adobe Creative Cloud WAM - End ##
10.180.216.92  shop.devcloud.local
```

The status bar at the bottom indicates 'Ln 26, Col 1', '963 characters', 'Plain text', '100%', 'Unix (LF)', and 'UTF-8'.



The screenshot shows a Windows PowerShell terminal window. The title bar says 'Windows PowerShell'. The output is as follows:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Brian ANTOINE> ping 10.180.216.92

Pinging 10.180.216.92 with 32 bytes of data:
Reply from 10.180.216.92: bytes=32 time=8ms TTL=64
Reply from 10.180.216.92: bytes=32 time=1ms TTL=64
Reply from 10.180.216.92: bytes=32 time=1ms TTL=64
Reply from 10.180.216.92: bytes=32 time=2ms TTL=64

Ping statistics for 10.180.216.92:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 8ms, Average = 3ms
PS C:\Users\Brian ANTOINE>
```


DevClo XDevClo XDevClo X

Not secure10.18...Summarize

Chat

 **DevCloud Shop**

Serveur : WEB1 | Session : 2l7ptbdoj8h29ub0dpbv1k8u1

 **Catalogue produits**

Produit	Prix	Action
Laptop	800 €	+ Ajouter
Phone	500 €	+ Ajouter
Headphones	100 €	+ Ajouter
Tablet	350 €	+ Ajouter

 **Panier**

Produit	Quantité	Sous-total	Action
Phone	1	500 €	Retirer
Headphones	1	100 €	Retirer
Total		600 €	

 **Valider la commande**

Total à payer : 600 €

Payer maintenant (simulation)



DevCloud Shop

Not secure shop.devcloud.local

Chat

 **DevCloud Shop**

Serveur : WEB1 | Session : ufrjgldgmiuamv9mquben9qe8

 **Catalogue produits**

Produit	Prix	Action
Laptop	800 €	+ Ajouter
Phone	500 €	+ Ajouter
Headphones	100 €	+ Ajouter
Tablet	350 €	+ Ajouter

 **Panier**

Votre panier est vide.

DevCloud Shop

DevCloud Shop

+

−

□

×

←↻⚠ Not secure shop.devcloud.local/index.php🔍☆👤⋮🗯 Chat

 **DevCloud Shop**

Serveur : WEB1 | Session : ufrjldgmuiamv9mquaben9qe8

 **Catalogue produits**

Produit	Prix	Action
Laptop	800 €	+ Ajouter
Phone	500 €	+ Ajouter
Headphones	100 €	+ Ajouter
Tablet	350 €	+ Ajouter

 **Panier**

Produit	Quantité	Sous-total	Action
Laptop	1	800 €	Retirer
Tablet	1	350 €	Retirer
Total		1150 €	

 **Valider la commande**

Total à payer : 1150 €

 Payer maintenant (simulation)